

SC2002 Group Assignment: Turn-Based Combat Arena (Java, OODP + SOLID)

1. Learning Outcomes

By completing this assignment, students will be able to:

- Design a non-trivial system using object-oriented programming principles.
- Apply SOLID design principles with concrete evidence.
- Document software design using UML class and sequence diagrams.
- Implement an extensible and testable Java system.

2. Game Overview

Build a turn-based combat arena game that runs entirely in the command line (no GUI).

For **experienced Java developers**, a GUI-based version may be implemented and will be accepted. However, **no additional marks will be awarded**, as this project places emphasis on **software design and object-oriented principles**, rather than user interface complexity. A player fights one or more enemies using Actions, Items, and Status Effects.

Game Flow

- The game proceeds in rounds.
- Each round, combatants act in a turn order determined by a TurnOrderStrategy.
- Player chooses an action via a text menu.
- Enemies perform BasicAttack.
- Win: all enemies defeated.
- Lose: player defeated.

Turn Order Strategy Flow

- In every game, there must be a clear ending, i.e. a winner and a loser
- There is no “draw” scenario

- In a turn, each entity takes one action. The order and sequence of actions depend on the speed stat of the entity. Higher speed goes first.

Logical Game Flow

Refer to the Appendix A: **Game Flow Examples**. Pay attention to the logic when each action is taken and their results. They show the important, non-trivial parts of your implementation.

Screen Display Details

<u>Loading Screen - Initiation</u> - List Players (Warrior and Wizard) - List Players' Attributes - Option to pick between Warrior and Wizard - Option to pick Item - List levels of difficulty – Easy, Medium, Hard (Explained in 3.5) - List number of enemy combatants by levels of difficulty - List Enemies (Goblin, Wolf) - List Enemies' Attributes - Option to select levels of difficulty
<u>During Gameplay</u> - Games proceed in round governed by TurnOrderStrategy - Conduct backup spawn of enemy combatants (if applicable) - Apply existing status effects first - Update HP (Existing HP-Damage+Defense), HP cannot go below 0, When damage reduces HP below 0, the system clamps it to 0. The character is then defeated / dead / eliminated. Check for game ending condition - Choose action (Player) - Action is executed - Update HP, status effects, inventory - End Turn, display alive/eliminated status - Check for game ending condition ➔ Players win if all enemies are defeated ➔ Players lose if HP reaches zero
<u>Game Completion Screen</u> - Player Victory Screen ➔ Congratulations, you have defeated all your enemies. ➔ Statistics: Remaining HP: Total Rounds: - Player Defeat Screen ➔ Defeated. Don't give up, try again! ➔ Statistics: Enemies remaining: Total Rounds Survived: ➔ Option to replay with the same settings, start a new game (return to the home screen), or exit.

3. Functional Requirements

- In your code, please follow the naming accordingly to help make our TAs work easier during the grading process
- Any details stated in the function requirements must be implemented in your final work for grading purposes
- We welcome self-initiated extra implementations, but no additional marks will be awarded for such implementation

3.1 Combatants

There are 2 Player classes – Warriors and Wizards.

There are 2 Enemy types – Goblin and Wolf.

Player	Attributes
Warrior	HP: 260 Attack: 40 Defense: 20 Speed: 30 Special Skill: Shield Bash Effect: Deal BasicAttack damage to selected enemy. Selected enemy is unable to take actions for the current turn and the next turn.
Wizard	HP: 200 Attack: 50 Defense: 10 Speed: 20 Special Skill: Arcane Blast Effect: Deal BasicAttack damage to all enemies currently in combat. Each enemy defeated by Arcane Blast adds 10 to the Wizard's Attack, lasting until end of the level.

Enemy	Attributes
Goblin	HP: 55 Attack: 35 Defense: 15 Speed: 25
Wolf	HP: 40 Attack: 45 Defense: 5 Speed: 35

3.2 Actions

Players can take 4 types of actions. Only one action can be taken in a turn.

Enemies can only perform BasicAttack.

Actions	Details
BasicAttack	- Performs an attack on a target selected by the user - $\text{Damage} = \max(0, \text{Attacker Attack} - \text{Target Defense})$ - Minimum HP post-damage is 0
Defend	- Increases defense by 10 for the current round and the next round
Item	- Uses the Item
SpecialSkill	- Execute class-specific special ability - Cooldown 3 turns, including current round - Track state for each combatant, decreasing cooldown only if a turn by the combatant took place

3.3 Status Effects

Below are the status effects:

Effect	Details
Arcane Blast	- Each enemy defeated by Arcane Blast adds 10 to the Wizard's Attack, lasting until end of the level.
Stun	- Affected entity is unable to take actions for the current turn and the next turn.

3.4 Items

- Two single-use items are chosen at game start, duplicates are allowed
- Items are used with an action

The items are as follows:

Items	Details
Potion	When used, Heal 100HP

	New HP= $\min(\text{Current HP} + \text{Heal Amount}, \text{Max HP})$
Power Stone	Trigger the special skill effect once, but it does not start or change the cooldown timer. In short, free extra use of the skill.
Smoke Bomb	When used, Enemy attacks do 0 damage in the current turn and the next turn

3.5 Level

There is a total of 3 levels that a player can challenge, ranging from Easy, to Medium, to Hard. Below are the specifications you can implement based on level no. in the game:

Level No.	Difficulty Level	Enemy Pool
1	Easy	Initial Spawn: 3 Goblins
2	Medium	Initial Spawn: 1 Goblin 1 Wolf Backup Spawn: 2 Wolf
3	Hard	Initial spawn: 2 Goblins Backup Spawn: 1 Goblin 2 Wolf

3.6 Advanced Logic

Backup Spawning

- Backup Spawns after the initial wave is completely defeated
- All entities of a Backup Spawn enter simultaneously

4. Architecture Requirements (IMPORTANT)

Your system should be designed using **object-oriented principles and appropriate abstractions**.

Students are expected to **identify suitable abstractions and responsibilities** when designing the system.

The architecture should support the following concepts:

i. **Battle Management**

The system should include a component responsible for:

- Managing the flow of battle rounds
- Determining the order in which characters take turns
- Processing actions performed during each turn

- Determining when the battle ends (e.g., when one side is defeated)

ii. Characters in Battle

The system should represent **different types of characters participating in battle**, such as players and enemies.

These characters should share common properties and behaviours, for example:

- Health points
- Combat statistics
- Ability to perform actions during their turn

Students should design appropriate abstractions to represent these combat participants.

iii. Combat Actions

Characters should be able to perform different actions during their turn, such as:

- Attacking
- Defending
- Using items
- Using special abilities

The system should support **extensibility**, allowing new actions to be added without major modification to existing code.

iv. Status Effects

Some actions may apply **effects that persist across multiple turns**, such as buffs or debuffs.

The system should support mechanisms to:

- Apply effects to characters
- Maintain effects across turns
- Remove effects when their duration ends

v. Usable Items

The game should include usable items that characters may use during battle. Each item may trigger different behaviours when used.

The design should allow **multiple item types** to exist.

vi. Turn Order Determination

The system should determine the **order in which characters act during each round**.

Extensibility: The system should be designed to allow different mechanisms for determining turn order to be supported in the future.**Enemy Behaviour**

vii. **Enemy Action:**

In this assignment, Enemies always execute **BasicAttack** on their turn. No behaviour variation or AI strategy is required.

Extensibility: The system should be designed so that different enemy action strategies can be supported in the future.

viii. **User Interface**

The user interface should be implemented as a **Command Line Interface (CLI)**.

The UI should:

- Display battle information
- Show available player actions
- Collect player input
- Pass user decisions to the battle system

The UI should be **separated from the battle logic**.

Design Justification Requirement

In the final submission, students must:

- Explain the **abstractions they designed**
- Justify their design decisions
- Describe how responsibilities are assigned to different components

A summary chart should be included explaining:

- Key abstractions
- Their responsibilities
- The design principles used (e.g., abstraction, encapsulation, polymorphism, design patterns).

5. SOLID Design Requirements

Students must explicitly demonstrate:

- SRP: Each class has one responsibility.
- OCP: New Action or StatusEffect added without modifying BattleEngine.
- LSP: Player and Enemy interchangeable as Combatant.
- ISP: No bloated interfaces.
- DIP: BattleEngine depends on abstractions, not concrete classes.

6. UML Deliverables

6.1 UML Class Diagram

Must show:

- Classes and interfaces
- Inheritance and associations
- Key attributes and methods
- Clear layering (UI, engine, domain)

6.2 UML Sequence Diagram

Model the following scenario accurately with the use of UML Sequence Diagram.

(Hint: You may design your own scenario or select one from the following options.)

1. Enemy's turn: Goblin executes basic attack on Warrior
2. Player's turn: Warrior defends against Goblin's next attack
3. Enemy's turn: Goblin executes basic attack, causing lower damage to Warrior
4. Player's turn: Warrior executes Shield Bash on Goblin, stunning it
5. Enemy's turn: Goblin is stunned, skips turn
6. Player's turn: Warrior uses Potion to heal
7. Enemy's turn: Goblin is stunned, skips turn
8. Player's turn: Warrior executes basic attack, eliminating Goblin

Please ensure the characters are big enough, and the lines are not intersecting one another.

Submission Requirements (Submit to the NTULearn lab group assignment submission link by the deadline: April 19, 11:59 PM.)

- Project report -one copy for whole group

THE REPORT

Your report will include the following:

a) A detailed UML **Class** Diagram for the application (Exported as an image and scaled up to page size.)

- Show clearly the class relationship, notation
- Notes to explain, if necessary
- Annotate your UML diagram to highlight where specific OO principles (e.g., encapsulation, polymorphism) are applied.

b) A detailed UML **Sequence** Diagram (Exported as an image and scaled up to page size.)

- The diagram should show clearly all participating objects involved with sufficient detailed flow and relevant interaction fragments.

c) Highlight clearly any **additional features/functionalities implemented in the system.**

d) Based on the concepts learned in the lecture, **write-up** on your **design considerations** and use of OO concepts in your current design, extensibility and maintainability of your design. Discuss any trade-offs you made in your design and reflect on how the design patterns you used contribute to the overall system design. Were there alternative patterns you considered? Why did you choose the ones you did?

e) Reflection: The difficulties encountered and the way to conquer, the knowledge learnt from this course, further improvement suggestions. Strong demonstration of learning points and insights of good design and implementation practices, based on experience gained from doing the assignment.

f) Include the link to your GitHub repository used for this project, containing all the relevant files and code.

- UML Class Diagram and Sequence Diagram (PDF or image with high resolution) -one copy for whole group
- Commit history showing individual contributions
- A duly signed **Declaration of Original Work** form. (Appendix B)
- A duly signed Declaration on Use of GAI Assistance for each member. (Appendix C)

ASSESSMENT WEIGHTAGE

UML Class Diagram [25 Marks]

- Show mainly the Entity classes, the essential Control and Boundary classes, and enumeration type (if there is).
- Clarity, Correctness and Completeness of details and relationship.

UML Sequence Diagram [20 Marks]

- Show only the sequence Diagram mentioned in 6.2
- Clarity, Correctness and Completeness of flow and object interactions details (Boundary-Control-Entities)

Design Consideration [15 Marks]

- Usage of OO concepts and principle - correctness and appropriateness
- Explanation of design choices and how it fits the project requirements
- Coupling and cohesion of classes

Implementation Code [20 Marks]

- Diagram to Code correctness, readability, Java naming convention, exception handling.
- Creativity of the additional features/functionality added to the system.

Demonstration and report [20 Marks]

- Coverage of application essentials and functionalities, user friendliness, demo flow, innovation.
- Report structure and reflection
- Highlight clearly any additional features implemented in the system.

Appendix A: Game Flow Examples

i. Difficulty Level (Easy)

Player: Warrior, **Warrior Stats:** HP: 260, ATK: 40, DEF: 20, SPD: 30

Items: Potion + Smoke Bomb

Level: Easy (3 Goblins) – A, B, C, **Goblin Stats:** HP: 55, ATK: 35, DEF: 15, SPD: 25

Turn Order: Warrior (SPD 30) → Goblins (SPD 25)

Round 1

Warrior → BasicAttack → Goblin A: HP: 55 → 30 (dmg: 40–15=25)

Goblin A → BasicAttack → Warrior: HP: 260 → 245 (dmg: 35–20=15)

Goblin B → BasicAttack → Warrior: HP: 245 → 230 (dmg: 35–20=15)

Goblin C → BasicAttack → Warrior: HP: 230 → 215 (dmg: 35–20=15)

End of Round 1: Warrior HP: 215/260 | Goblin A HP: 30 | Goblin B HP: 55 | Goblin C HP: 55 |
Potion: 1 | Smoke Bomb: 1 | Special Skills Cooldown: 0 Round

Round 2

Warrior → Shield Bash → Goblin A: HP: 30 → 5 (dmg: 40–15=25) | Goblin A STUNNED (2 turns)

Goblin A → STUNNED: Turn skipped

Goblin B → BasicAttack → Warrior: HP: 215 → 200 (dmg: 35–20=15)

Goblin C → BasicAttack → Warrior: HP: 200 → 185 (dmg: 35–20=15)

End of Round 2: Warrior HP: 185/260 | Goblin A HP: 5 [STUNNED] | Goblin B HP: 55 |
Goblin C HP: 55 | Potion: 1 | Smoke Bomb: 1 | Special Skills Cooldown: 3 rounds

Round 3

Warrior → BasicAttack → Goblin A: HP: 5 → 0 X ELIMINATED (dmg: 40–15=25)

Goblin A → ELIMINATED: Skipped | Stun expires

Goblin B → BasicAttack → Warrior: HP: 185 → 170 (dmg: 35–20=15)

Goblin C → BasicAttack → Warrior: HP: 170 → 155 (dmg: 35–20=15)

End of Round 3: Warrior HP: 155/260 | Goblin A HP: X | Goblin B HP: 55 | Goblin C HP: 55 |
Potion: 1 | Smoke Bomb: 1 | Special Skills Cooldown: 2 rounds

Round 4

Warrior → Item → Smoke Bomb used: Enemy attacks deal 0 damage this turn + next

Goblin B → BasicAttack → Warrior: 0 damage (Smoke Bomb active) | Warrior HP: 155

Goblin C → BasicAttack → Warrior: 0 damage (Smoke Bomb active) | Warrior HP: 155

End of Round 4: Warrior HP: 155/260 | Goblin B HP: 55 | Goblin C HP: 55 | Potion: 1 | Smoke Bomb: 0 ← consumed | Effect: 1 turn remaining | Special Skills Cooldown: 1 round

Round 5

Warrior → Shield Bash → Goblin B: HP: 55 → 30 (dmg: 40–15=25) | Goblin B STUNNED (2 turns)

Goblin B → BasicAttack → Warrior: 0 damage (Smoke Bomb last use) | Smoke Bomb effect expires | Warrior HP: 155

Goblin C → BasicAttack → Warrior: HP: 155 → 140 (dmg: 35–20=15)

End of Round 5: Warrior HP: 140/260 | Goblin B HP: 30 [STUNNED] | Goblin C HP: 55 |
Potion: 1 | Smoke Bomb: 0 | Special Skills Cooldown: 3 rounds

Round 6

Warrior → BasicAttack → Goblin C: HP: 55 → 30 (dmg: 40–15=25)

Goblin B → STUNNED: Turn skipped | Stun expires

Goblin C → BasicAttack → Warrior: HP: 140 → 125 (dmg: 35–20=15)

End of Round 6: Warrior HP: 125/260 | Goblin B HP: 30 | Goblin C HP: 30 | Potion: 1 | Smoke Bomb: 0 | Special Skills Cooldown: 2 rounds

Round 7

Warrior → **Item** → **Potion used**: HP: 125 → 225 (+100)

Goblin B → **BasicAttack** → **Warrior**: HP: 225 → 210 (dmg: 35-20=15)

Goblin C → **BasicAttack** → **Warrior**: HP: 210 → 195 (dmg: 35-20=15)

End of Round 7: Warrior HP: 195/260 | Goblin B HP: 30 | Goblin C HP: 30 | Potion: 0 ← consumed | Smoke Bomb: 0 | Item action no longer available | Special Skills Cooldown: 1 round

Round 8

Warrior → **Shield Bash** → **Goblin B**: HP: 30 → 5 (dmg: 40-15=25) | Goblin B STUNNED (2 turns)

Goblin B → **STUNNED**: Turn skipped

Goblin C → **BasicAttack** → **Warrior**: HP: 195 → 180 (dmg: 35-20=15)

End of Round 8: Warrior HP: 180/260 | Goblin B HP: 5 [STUNNED] | Goblin C HP: 30 | Potion: 0 | Smoke Bomb: 0 | Special Skills Cooldown: 3 rounds

Round 9

Warrior → **BasicAttack** → **Goblin B**: HP: 5 → 0 X ELIMINATED (dmg: 40-15=25) | Cooldown: 2

Goblin B → **ELIMINATED**: Skipped | Stun expires

Goblin C → **BasicAttack** → **Warrior**: HP: 180 → 165 (dmg: 35-20=15)

End of Round 9: Warrior HP: 165/260 | Goblin B HP: 0 | Goblin C HP: 30 | Potion: 0 | Smoke Bomb: 0 | Special Skills Cooldown: 2 rounds

Round 10

Warrior → **BasicAttack** → **Goblin C**: HP: 30 → 5 (dmg: 40-15=25)

Goblin C → **BasicAttack** → **Warrior**: HP: 165 → 150 (dmg: 35-20=15)

End of Round 10: Warrior HP: 150/260 | Goblin C HP: 5 | Potion: 0 | Smoke Bomb: 0 | Special Skills Cooldown: 1 rounds

Round 11

Warrior → **BasicAttack** → **Goblin C**: HP: 5 → 0 ✗ ELIMINATED (dmg: 40-15=25) | All enemies defeated

End of Round 11: Warrior HP: 150/260 | Goblin A HP: 0 | Goblin B HP: 0 | Goblin C HP: 0 |
Potion: 0 | Smoke Bomb: 0 | Special Skills Cooldown: 0 rounds

Victory

Result: Player Victory **Remaining HP**: 150 / 260 | **Total Rounds**: 11 | **Remaining Potion**: 0 |
Remaining Smoke Bomb: 0

ii. Difficulty Level (Medium)

Player: Warrior, **Player Stats**: HP: 260, ATK: 40, DEF: 20, SPD: 30

Items: Power Stone + Potion

Level: Medium - 1 Goblin + 1 Wolf | Backup: Wolf A + Wolf B

Goblin Stats: HP: 55, ATK: 35, DEF: 15, SPD: 25

Wolf Stats: HP: 40, ATK: 45, DEF: 5, SPD: 35

Turn Order: Wolf (SPD 35) → Warrior (SPD 30) → Goblin (SPD 25)

Round 1

Wolf → **BasicAttack** → **Warrior**: HP: 260 → 235 (dmg: 45-20=25)

Warrior → **Shield Bash** → **Wolf**: HP: 40 → 5 (dmg: 40-5=35) | Wolf STUNNED (2 turns) |
Cooldown set to 3

Goblin → **BasicAttack** → **Warrior**: HP: 235 → 220 (dmg: 35-20=15)

End of Round 1: Warrior HP: 220/260 | Goblin HP: 55 | Wolf HP: 5 [STUNNED] | Power
Stone: 1 | Potion: 1 | Special Skills Cooldown: 3 rounds

Round 2

Wolf → **STUNNED**: Turn skipped

Warrior → **BasicAttack** → **Wolf**: HP: 5 → 0 ✗ ELIMINATED (dmg: 40-5=35)

Wolf → **ELIMINATED**: Skipped | Stun expires

Goblin → **BasicAttack** → **Warrior**: HP: 220 → 205 (dmg: 35–20=15)

End of Round 2: Warrior HP: 205/260 | Goblin HP: 55 | Wolf HP: 0 | Power Stone: 1 | Potion: 1
| Special Skills Cooldown: 2 rounds

Round 3

Turn Order: Warrior (SPD 30) → Goblin (SPD 25)

Warrior → **BasicAttack** → **Goblin**: HP: 55 → 30 (dmg: 40–15=25)

Goblin → **BasicAttack** → **Warrior**: HP: 205 → 190 (dmg: 35–20=15)

End of Round 3: Warrior HP: 190/260 | Goblin HP: 30 | Wolf HP: 0 | Power Stone: 1 | Potion: 1
| Special Skills Cooldown: 1 round

Round 4

Warrior → **BasicAttack** → **Goblin**: HP: 30 → 5 (dmg: 40–15=25)

Goblin → **BasicAttack** → **Warrior**: HP: 190 → 175 (dmg: 35–20=15)

End of Round 4: Warrior HP: 175/260 | Goblin HP: 5 | Wolf HP: 0 | Power Stone: 1 | Potion: 1
Special Skills Cooldown: 0 rounds

Round 5

Warrior → **Shield Bash** → **Goblin**: HP: 5 → 0 ✗ **ELIMINATED** (dmg: 40–15=25) | Cooldown set to 3

Goblin → **ELIMINATED**: Skipped

All initial enemies eliminated → **Backup Spawn triggered!** Wolf A (HP: 40) + Wolf B (HP: 40) enter simultaneously

End of Round 5: Warrior HP: 175/260 | Goblin HP: 0 | Wolf HP: 0 | Wolf A HP: 40 | Wolf B HP: 40 | Power Stone: 1 | Potion: 1 | Special Skills Cooldown: 3 rounds

Round 6

Turn Order: Wolf A (SPD 35) → Wolf B (SPD 35) → Warrior (SPD 30)

Wolf A → BasicAttack → Warrior: HP: 175 → 150 (dmg: 45–20=25)

Wolf B → BasicAttack → Warrior: HP: 150 → 125 (dmg: 45–20=25)

Warrior → Item → Power Stone used → Shield Bash triggered → Wolf A: Wolf A HP: 40 → 5 (dmg: 40–5=35) | Wolf A STUNNED (2 turns) | Cooldown unchanged → 3 (Power Stone does not affect cooldown) | Power Stone consumed

End of Round 6: Warrior HP: 125/260 | Wolf A HP: 5 [STUNNED] | Wolf B HP: 40 | Power Stone: 0 ← consumed | Potion: 1 | Special Skills Cooldown: 3 rounds

Round 7

Wolf A → STUNNED: Turn skipped

Wolf B → BasicAttack → Warrior: HP: 125 → 100 (dmg: 45–20=25)

Warrior → BasicAttack → Wolf A: HP: 5 → 0 X ELIMINATED (dmg: 40–5=35)

Wolf A → ELIMINATED: Skipped | Stun expires

End of Round 7: Warrior HP: 100/260 | Wolf A HP: 0 | Wolf B HP: 40 | Power Stone: 0 | Potion: 1 | Special Skills Cooldown: 2 rounds

Round 8

Wolf B → BasicAttack → Warrior: HP: 100 → 75 (dmg: 45–20=25)

Warrior → BasicAttack → Wolf B: HP: 40 → 5 (dmg: 40–5=35)

End of Round 8: Warrior HP: 75/260 | Wolf A HP: 0 | Wolf B HP: 5 | Power Stone: 0 | Potion: 1 | Special Skills Cooldown: 1 round

Round 9

Wolf B → BasicAttack → Warrior: HP: 75 → 50 (dmg: 45–20=25)

Warrior → BasicAttack → Wolf B: HP: 5 → 0 X ELIMINATED (dmg: 40–5=35) | All enemies defeated

End of Round 9: Warrior HP: 50/260 | Wolf A HP: 0 | Wolf B HP: 0 | Power Stone: 0 | Potion: 1 ← unused | Special Skills Cooldown: 0 rounds

Victory

Result: Player Victory **Remaining HP:** 50/260 | **Total Rounds:** 9 | **Remaining Power Stone:** 0 | **Remaining Potion:** 1 ← unused

iii. Difficulty Level (Medium)

Player: Wizard, **Player Stats:** HP: 200, ATK: 50, DEF: 10, SPD: 20

Items: Power Stone + Potion

Level: Medium - 1 Goblin + 1 Wolf | Backup: Wolf A + Wolf B

Goblin Stats: HP: 55, ATK: 35, DEF: 15, SPD: 25

Wolf Stats: HP: 40, ATK: 45, DEF: 5, SPD: 35

Turn Order: Wolf (SPD 35) → Goblin (SPD 25) → Wizard (SPD 20)

Round 1

Wolf → BasicAttack → Wizard: HP: 200 → 165 (dmg: 45-10=35)

Goblin → BasicAttack → Wizard: HP: 165 → 140 (dmg: 35-10=25)

Wizard → Arcane Blast → All Enemies: Wolf HP: 40 → 0 X ELIMINATED (dmg: 50-5=45) | ATK: 50 → 60 (+10 per Arcane Blast kill) Goblin HP: 55 → 20 (dmg: 50-15=35) | Goblin survives | Cooldown set to 3

End of Round 1: Wizard HP: 140/200 | Goblin HP: 20 | Wolf HP: 0 | Power Stone: 1 | Potion: 1 | Special Skills Cooldown: 3 rounds | Wizard ATK: 60

Round 2

Turn Order: Goblin (SPD 25) → Wizard (SPD 20)

Goblin → BasicAttack → Wizard: HP: 140 → 115 (dmg: 35-10=25)

Wizard → BasicAttack → Goblin: HP: 20 → 0 X ELIMINATED (dmg: 60-15=45) | ATK bonus applies to all attacks for rest of level

Goblin → ELIMINATED: Skipped

All initial enemies eliminated → **Backup Spawn triggered!** Wolf A (HP: 40) + Wolf B (HP: 40) enter simultaneously

End of Round 2: Wizard HP: 115/200 | Goblin HP: 0 | Wolf HP: 0 | Wolf A HP: 40 | Wolf B HP: 40 | Power Stone: 1 | Potion: 1 | Special Skills Cooldown: 2 rounds | Wizard ATK: 60

Round 3

Turn Order: Wolf A (SPD 35) → Wolf B (SPD 35) → Wizard (SPD 20)

Wolf A → BasicAttack → Wizard: HP: 115 → 80 (dmg: 45-10=35)

Wolf B → BasicAttack → Wizard: HP: 80 → 45 (dmg: 45-10=35)

Wizard → Item → Power Stone used → Arcane Blast triggered → All Enemies (ATK: 60):
Wolf A HP: 40 → 0 X ELIMINATED (dmg: 60-5=55) | ATK: 60 → 70 (+10) Wolf B HP: 40 → 0 X ELIMINATED (dmg: 60-5=55) | ATK: 70 → 80 (+10) Power Stone consumed | Cooldown unchanged → 2 (Power Stone does not affect cooldown) | All enemies defeated

End of Round 3: Wizard HP: 45/200 | Wolf A HP: 0 | Wolf B HP: 0 | Power Stone: 0 ← consumed | Potion: 1 ← unused | Special Skills Cooldown: 2 rounds | Wizard ATK: 80

Victory

Result: Player Victory **Remaining HP:** 45/200 | **Total Rounds:** 3 | **Remaining Power Stone:** 0 | **Remaining Potion:** 1 ← unused | **Final Wizard ATK:** 80

Appendix B

Declaration of Original Work for SC2002/CE2002/CZ2002 Assignment

We hereby declare that the attached group assignment has been researched, undertaken, completed, and submitted as a collective effort by the group members listed below.

We have honored the principles of academic integrity and have upheld Student Code of Academic Conduct in the completion of this work.

We understand that if plagiarism is found in the assignment, then lower marks or no marks will be awarded for the assessed work. In addition, disciplinary actions may be taken.

Name	Course (SC2002/CE2002 CZ2002)	Lab Group	Signature /Date

Important notes:

1. Name must **EXACTLY MATCH** the one printed on your Matriculation Card.
2. Student Code of Academic Conduct includes the latest guidelines on usage of Generative AI and any other guidelines as released by NTU.

Appendix C:

Declaration on Use of GAI (Generative Artificial Intelligence) Assistance in relation to Assignment/Project (to be submitted individually even for group projects)

I _____ (student name),
_____@e.ntu.edu.sg (NTU email) honestly and sincerely make the following
declaration in relation to the following course submission:

1. Name of course:
2. Course Code:
3. Instructor:
4. Title of Assignment/Project Submission:

In relation to the foregoing I hereby declare that, fully and properly in accordance with the Assignment/Project Instructions I have (check where appropriate):

- i. Used GAI as permitted to assist in generating key ideas only. ☐
- ii. Used GAI as permitted to assist in generating a first text only. ☐

And/or

- iii. Used GAI to refine syntax and grammar for correct language submission only.

☐

Or

- iv. As it is not permitted: Not used GAI assistance in any way in the development or generation of this assignment or project.

☐

I also declare that I have :

- a. Fully and honestly submitted the digital paper trail required under the assignment/project instructions; and that

b. Wherever GAI assistance has been employed in the submission in word or paraphrase or inclusion of a significant idea or fact suggested by the GAI assistant, I have acknowledged this by a footnote; and that,

c. Apart from the foregoing notices, the submission is wholly my own work.

Student Name & Signature Date

(Optional) Sample Reflection Questions:

In 150 words or less, briefly state up to 5 key ideas that you formulated that the GAI tools alone could not have generated.